



Multidisciplinary tumor board decisions and artificial intelligence–generated recommendations in general surgery: a retrospective observational study

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Abstract

Artificial intelligence (AI)–based decision support systems are increasingly explored in surgical oncology. However, their concordance with multidisciplinary tumor board (MTB) decisions, particularly in complex gastrointestinal malignancies, remains insufficiently characterized. This retrospective observational study included 47 patients who underwent oncological surgery and were discussed at a multidisciplinary tumor board during an eighteen-month period. For each patient, concordance between MTB-established treatment decisions and subsequent AI-generated recommendations (ChatGPT-based) was assessed. Concordance was categorized as discordant (0), partially concordant (1), or fully concordant (2). Discordant cases were further analyzed across predefined domains, including staging discrepancies, resectability assessment, interpretation of metastatic disease burden, and treatment sequencing. Full concordance was observed in 18 cases (38.3%), partial concordance in 22 cases (46.8%), and discordance in 7 cases (14.9%). Overall, 85.1% of cases demonstrated at least partial concordance. Discordance was more frequent among male patients (5 of 19; 26.3%) compared with female patients (2 of 28; 7.1%). Qualitative analysis revealed that discordance most commonly arose from differences in resectability assessment and staging interpretation. AI recommendations tended to favor broader surgical candidacy, whereas MTB decisions more frequently excluded surgery based on nuanced clinical and contextual factors. AI-generated recommendations show substantial overlap with MTB decisions in most cases but diverge at critical surgical decision points, particularly regarding resectability and staging. These findings suggest that AI may serve as a complementary decision-support tool rather than a substitute for multidisciplinary clinical judgment for oncological surgery care.

Keywords ChatGPT · Surgical oncology · Large language models · Clinical decision support · Treatment concordance

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Introduction

The development of novel therapies, ongoing revisions of clinical guidelines, and the rapid evolution of scientific evidence make treatment planning for patients with cancer increasingly complex [1]. Multidisciplinary tumor boards (MTBs) are widely regarded as a fundamental element of modern oncologic decision-making, especially in surgical oncology, where therapeutic strategies require careful coordination and input from multiple specialties [2]. Treatment strategies rely on detailed evaluations of resectability, disease stage, existing comorbid conditions, and projected operative risk. In this context, MTBs not only contribute to the development of individualized treatment strategies through comprehensive multidisciplinary evaluation, but may also improve the overall quality and coordination of oncological care by facilitating accurate staging, adherence to clinical guidelines, timely treatment planning, and potentially better clinical outcomes [3–5].

The rapid advancement of artificial intelligence across diverse fields has also been reflected in healthcare delivery, paving the way for the development of clinical decision support systems. This evolution contributes to improved management of oncology patients in terms of quality of life, while also helping to reduce overall healthcare costs [6]. Preliminary evaluations indicate that artificial intelligence (AI) demonstrates a high level of concordance with the decisions made by expert MTBs, while also reducing cognitive workload [1]. Reported rates of AI–MTB agreement across different cancer types range between 70 and 90% [7]. Most existing studies are primarily based on simulated or relatively homogeneous datasets, which restricts the generalizability of their findings to the diverse and complex populations encountered in surgical oncology [8, 9]. These gaps underscore the need for pragmatic assessments of AI performance within real-world MTB settings, thereby providing a clear rationale for systematically comparing AI-generated recommendations with multidisciplinary tumor board decisions in everyday clinical practice.

The present study aims to evaluate the degree of concordance between MTB decisions and AI-based recommendations in oncological surgery, and to analyze the domains in which discordance occurs.

Materials and methods

Study design and patient population

This retrospective observational study included patients discussed at the multidisciplinary oncology council of a tertiary referral center during June 2023 to December 2024

(Ethical approval number: 2025–03). The MTB consists of an oncologist, radiologist, radiation oncologist, nuclear medicine specialist, pathologist, and general surgeon. In our institution, MTB discussions are primarily conducted for patients with complex oncologic cases requiring multidisciplinary evaluation and treatment planning, rather than for all patients with a cancer diagnosis. Therefore, a potential selection bias should be acknowledged. The cohort intentionally included a heterogeneous spectrum of malignancies, including colorectal, gastric, hepatopancreatobiliary, and ovarian cancers. Accordingly, the presented multidisciplinary discussion structure should be interpreted as one possible institutional model reflecting routine clinical practice, rather than a universal or tumor-specific MTB framework. Patients with incomplete clinical, pathological, or radiological data, those without a confirmed pathological cancer diagnosis, cases presented only for radiological evaluation were excluded.

Multidisciplinary tumor board decision

For each patient, the final MTB decision recorded during the tumor board meeting was considered the reference standard. Decisions included surgical management, neoadjuvant or adjuvant therapy, non-surgical approaches, or palliative strategies.

AI-based evaluation

For each case, ChatGPT-4.0 (OpenAI, San Francisco, CA, USA) was consulted retrospectively for exploratory purposes after completion of the MTB decisions. The same model version was used consistently throughout the study period. Queries were conducted in Turkish using a standardized prompt structure to minimize variability in AI-generated responses. Each query began with the instruction: “Evaluate the following oncologic case according to current evidence-based guidelines and provide recommendations regarding diagnosis, staging, and treatment planning.” The AI system was then provided with anonymized patient data, including age, sex, tumor type, TNM staging, imaging findings (reported in text form), pathological results, relevant comorbidities, and the specific clinical question regarding diagnosis or treatment planning. Importantly, the AI system was blinded to MTB decisions and generated recommendations solely based on the provided clinical information and guideline-based reasoning (Fig. 1).

Concordance assessment

Concordance between MTB recommendations and ChatGPT-4.0 outputs was independently evaluated by two

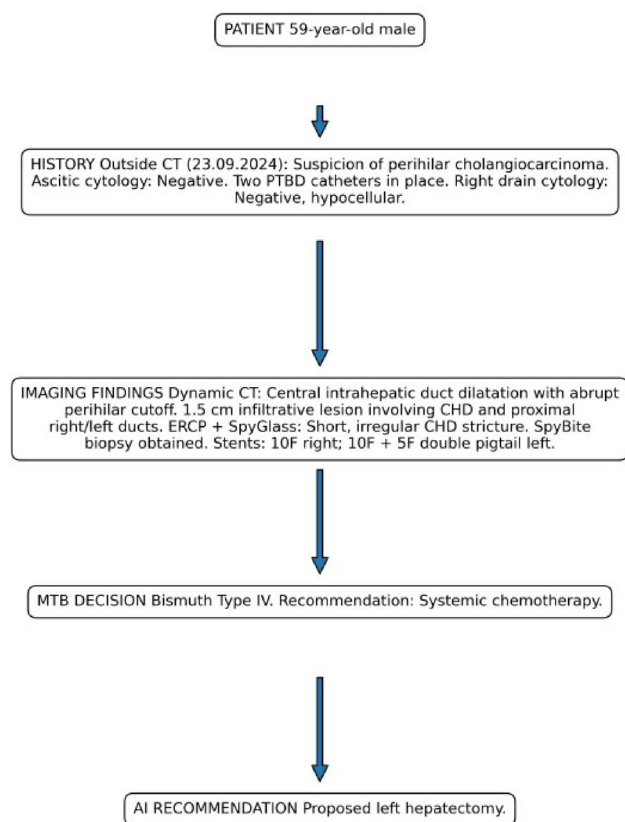


Fig. 1 Visual summary of diagnostic findings and treatment recommendations (MTB vs AI) in a tumor board patient

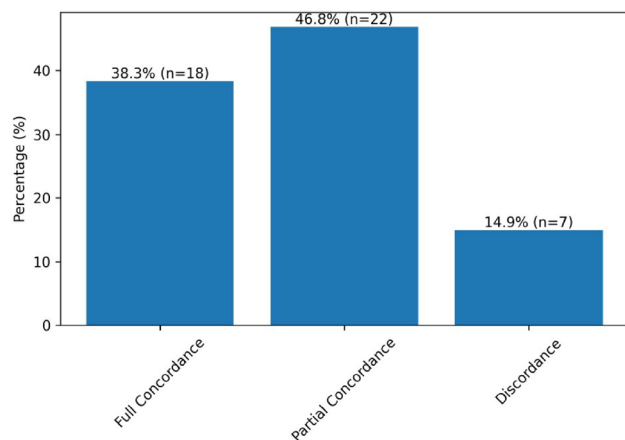


Fig. 2 Overall concordance between MTB and AI recommendations

general surgeons with experience in oncologic multidisciplinary management. Final categorization was determined by consensus discussion between the reviewers. Inter-rater agreement between the two surgeons was assessed using Cohen's kappa coefficient. The analysis demonstrated substantial agreement ($\kappa=0.63$), with an observed agreement of 76.6%.

Concordance was classified into three categories: complete concordance, partial concordance, and discordance. Complete concordance was defined as substantial agreement between the MTB and AI recommendations regarding diagnostic interpretation, staging assessment, and treatment strategy. Partial concordance referred to cases in which either the clinical reasoning process was appropriate but the final recommendation differed, or the final recommendation was similar despite differences in the proposed reasoning or management pathway. Discordance was defined as clinically meaningful disagreement between the MTB and AI recommendations with respect to diagnosis, staging, or treatment planning.

Representative examples of partial concordance included cases in which the AI suggested an appropriate multimodal treatment approach but recommended a different treatment sequence, or cases in which the final therapeutic recommendation was similar to the MTB decision despite differing clinical rationale.

Discordant cases were further categorized into predefined domains: staging discrepancy, resectability assessment, metastatic interpretation, and treatment sequencing.

Results

Overall concordance

A total of 47 out of 212 consulted patients were evaluated within the scope of the oncologic surgery department. Among the 47 patients, full concordance between MTB and AI recommendations was observed in 18 cases (38.3%). Partial concordance was identified in 22 cases (46.8%), while 7 cases (14.9%) were classified as discordant. Thus, 85.1% of cases demonstrated at least partial alignment between MTB and AI recommendations (Fig. 2).

Concordance by sex

Discordance was more frequent among male patients (5 of 19; Qualitative analysis revealed that discordance most commonly 26.3%) compared with female patients (2 of 28; 7.1%). Partial and full concordance predominated in female patients (Fig. 3).

Nature of discordance

Qualitative analysis of discordant cases revealed that discrepancies most commonly involved:

- *Resectability assessment*: AI more frequently identified potential surgical eligibility in cases where MTB

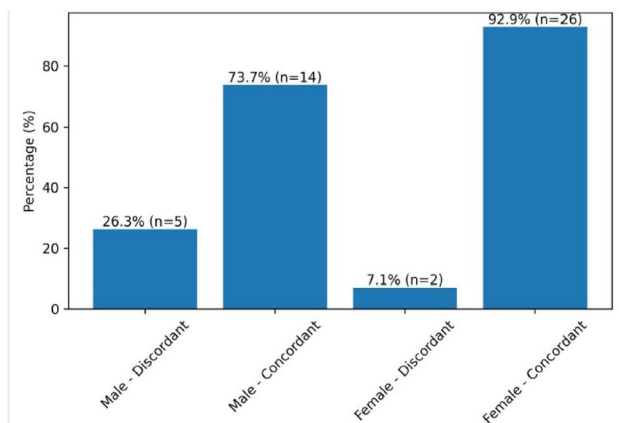


Fig. 3 Concordance and discordance by sex

decisions excluded surgery based on anatomical extent, prior treatments, or institutional practices (n=2).

- *Staging interpretation:* Differences in interpretation of tumor extent or metastatic burden led to divergent treatment strategies (n=3).
- *Treatment sequencing:* Divergence in the proposed order of surgery, chemotherapy, and radiotherapy (n=2).

Discussion

In this study, full concordance between MTB decisions and ChatGPT-based AI recommendations was observed in 38.3% of cases, partial concordance in 46.8%, and discordance in 14.9%, yielding an overall rate of at least partial agreement of 85.1%. These findings suggest that large language model-based AI systems may be able to approximate the fundamental treatment directions adopted by MTBs in routine surgical oncology practice; however, the results should be interpreted with caution given the limited sample size and the monocentric design of the study. Nevertheless, clinically relevant discrepancies emerged at key surgical decision points, most notably in assessments of resectability and disease staging. In these situations, AI more frequently suggested broader surgical eligibility, whereas MTB decisions were shaped by individualized clinical judgment that incorporated patient-specific factors, prior treatments, and cumulative comorbidity burden. Discordant decisions highlighted that MTB evaluations were informed not only by clinical data but also by experiential judgment and broader contextual factors.

One of the major strengths of this study lies in its inclusion of a heterogeneous surgical oncology population. Rather than focusing on a single tumor type, this cohort reflects the diversity encountered in real-life MTBs. Although concordance varied across surgical domains, this heterogeneity enhances the external validity of the findings

and underscores the importance of MTB oversight when AI tools are applied across different oncologic contexts. Crucially, the present findings do not support the notion that AI can replace multidisciplinary oncology councils. Instead, they highlight the potential role of AI as a complementary decision-support tool, particularly in structuring treatment options, summarizing clinical information, and identifying guideline-consistent strategies.

The nature of discordance, particularly in tumor extent and metastatic burden assessment, may be influenced by variability in radiological interpretation, including differences in imaging acquisition protocols and reporting styles across institutions. However, in the present study, radiological reports were not stratified according to whether imaging was performed internally or externally. In addition, the AI system was provided only with the written CT reports and did not have access to the real-time interpretations or additional contextual input provided by radiologists during MTB discussions. Such differences in available radiological information may have contributed to divergent interpretations between MTB decisions and AI-generated recommendations. Another results of the study is the observed higher discordance rate among male patients may reflect differences in case complexity or tumor characteristics within this subgroup; however, given the limited sample size, this finding should be interpreted cautiously and cannot be considered definitive.

Large language models have demonstrated strong performance in addressing medical inquiries and outlining clinical protocols [10]. Multidisciplinary decisions integrate patient-specific considerations and local institutional practices that cannot be entirely reflected by artificial intelligence systems predominantly grounded in guideline-based reasoning [11]. In our study, although the overall concordance rate was high, discordance most frequently emerged at critical decision thresholds, particularly in assessments of resectability and disease staging. These domains represent the most complex and experience-dependent components of surgical oncology.

In cases of disagreement, a consistent pattern emerged in which AI systems were more likely than MTBs to designate patients as suitable candidates for surgery. This inclination may stem from the AI's dependence on guideline-driven frameworks and idealized clinical scenarios. In a prospective study, treatment decisions made by multidisciplinary tumor boards were compared with recommendations generated by the large language model ChatGPT. Among the 100 patients evaluated, the overall concordance rate was found to be 76.4%. Discrepancies most frequently arose in cases where patient-specific individual factors were not adequately taken into account [1]. In a retrospective study evaluating 322 cases of gastric cancer, the concordance rate

was reported as 86.96%. Factors associated with agreement included patient age, performance status, and a diagnosis of stage IV gastric cancer [12]. From a clinical perspective, the observed discordance emphasizes that AI systems currently lack access to several intangible but critical elements of MDT decision-making, including intraoperative experience, institutional protocols, and patient preferences [11]. Ethically, this reinforces the responsibility of clinicians to critically appraise AI-generated suggestions rather than adopting them uncritically.

One reason for the lack of clarity in explaining the results of artificial intelligence is that different forms of expression elicit different responses. Therefore, standardization of the process is crucial [9]. In a study, the clinical outcomes of patients diagnosed with sarcoma were each queried 21 times across various artificial intelligence applications. Even the best-performing system achieved concordance with expert consensus in only 60% of cases. The analysis revealed substantial differences between large language models and human specialists, with large language models occasionally generating alternative recommendations that could be potentially harmful. Moreover, variability among models in terms of guideline adherence and source utilization underscored the importance of integrating region-specific clinical standards into AI training processes [11]. Another significant limitation of model-generated recommendations is the lack of transparency. ChatGPT does not clearly disclose the specific data sources or evidence underpinning its suggestions. This opacity may create a safety gap for clinicians, particularly when making patient-specific and high-stakes clinical decisions [1]. For effective implementation, these systems should be embedded within hospital information infrastructures using architectures compatible with electronic health record systems and interoperable application programming interfaces [7].

Important limitations of this study include its single-center design and relatively small sample size, which may limit the generalizability of findings. AI recommendations were generated retrospectively, excluding real-time MDT discussions and patient preferences, which are integral to actual clinical decision-making. Variation in AI configuration, prompt construction, and model version could influence recommendation quality, and these factors may differ across different AI systems and clinical implementations.

Conclusions

AI-generated recommendations demonstrate substantial concordance with MDT decisions but show meaningful divergence at key surgical decision points. AI should be regarded as a decision-support tool rather than a replacement

for multidisciplinary clinical judgment. In complex surgical oncology cases, human expertise remains indispensable. This study illustrates not what AI can do, but where it should stop.

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Declarations

Conflict of interest The authors declared no conflict of interest.

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